When should a neural text-entry decoder commit? Risk-controlled commits from a stream-level trust layer

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Abstract

Objective. Neural text-entry decoders often end in a language model that writes fluent sentences from weak neural evidence. We ask when such a system should send (*commit*) a sentence without confirmation, and whether wrong commits can be bounded. *Approach.* A trust layer reads the decoder's per-frame output stream, calibrates its confidence, and commits, offers alternatives or abstains. A Learn-then-Test threshold, never below a speed-tuned threshold, bounds the wrong-commit rate with high probability. The rule was chosen after the random re-splits below; all inference is exploratory. We test it offline on one intracortical speech participant (Brain-to-Text '25, 14 days; our decoder, and the published sentence gated by agreement with it), calibrating before testing in time. Forearm sEMG and MEG typing serve as format and safety checks. *Main results.* Always committing sent wrong text for 61% and 19% of sentences. Calibrated on earlier days, the rule's pooled wrong-commit rate was 1.0–2.7% (target 5%); no day was significantly above 5%. Over 50 random re-splits, 8% (95% CI 2–19%) and 2% (0.1–11%) exceeded 5%; the rule committed 15% and 39% of sentences, and 21% and 8% of commits were wrong. The bound covers the mix of days: one day ran at 10%. On our decoder, a plain threshold committed 198 of 300 no-information streams; with an evidence guard, none. *Significance.* A stream-level commit rule kept wrong commits rare across days for one participant. It needs an input-validity check and per-session calibration, and commits few sentences with current decoders. *Clinical trial registration:* BrainGate2, ClinicalTrials.gov NCT00912041.

Keywords: brain–computer interface, text entry, language model, risk control, Learn then Test, selective prediction, safety

1 Introduction

Brain–computer interfaces (BCIs) for text entry are moving towards clinical and consumer use. Intracortical neuroprostheses decode attempted speech, handwriting and typing at high rates [1–5]; non-invasive decoders reconstruct typed sentences from magnetoencephalography (MEG) [6, 7]; and surface electromyography (sEMG) is a candidate input for wearables [8–10]. Most of these systems end in a language-model (LM) stage that turns noisy per-frame predictions into sentences. This lowers the error rate but changes the kind of error: when the neural evidence is weak, the LM still returns a grammatical sentence, and a reader has no cue that it is wrong. In speech recognition, such fluent unsupported output has caused documented harm [11].

For people who depend on the interface to communicate, accuracy is a stated priority [12–15], and users of augmentative and alternative communication (AAC) want LM-generated text to reflect their own intent [16, 17]. A sentence sent in their name that they did not mean is a harm in itself. Current speech neuroprostheses read a finalised sentence aloud without a confirmation step [2] or synthesise voice instantaneously [18]. Whether a decoded sentence should be sent without confirmation (*committed*) is therefore a real design decision.

Sentence-level LM decoders for speech, handwriting, typing and sEMG share one structure: per-frame probabilities followed by an LM search. A commit rule can therefore act on the

per-frame output instead of inside one decoder. The statistical tools exist: calibration, selective classification, conformal risk control (CRC) and Learn-then-Test (LTT) [19–23]. They have been applied to single BCI decoders [24, 25] and to speech recognition [26]. The methods are not new. What is missing is their use as a commit rule for neural text entry, a test of the guarantee when calibration comes before use in time, and the fair comparators: "always offer, never commit" and a no-BCI baseline such as gaze typing [27].

This work. We define the problem (Section 3), a trust layer on a per-frame output stream (Section 4) and an evaluation protocol (Section 5), and test them offline on public data. Our contributions are:

1. **A stream-level commit rule with a stated guarantee.** The layer calibrates the per-frame log-probabilities and chooses COMMIT, OFFER or ABSTAIN. LTT with a max rule bounds the wrong-commit rate with high probability (Algorithm 1; Algorithm S1 in the supplement).
2. **A time-ordered test of the guarantee on one participant.** We calibrate on earlier days or on the start of a session and test on later data, over 14 days. We also show the scope of the guarantee: it bounds the risk over the mix of days, not on each day.
3. **A safety-first evaluation protocol**: the wrong-commit rate with the commit rate and the error among commits; always-commit, always-offer and gaze-typing baselines; a simulated-user speed metric with a sensitivity analysis; and signal-destroying controls. These controls show that a plain confidence threshold commits no-information streams, so a commit rule needs an input-validity check.

The stream format (NeuralTextStream v0.1), its adapters, an audit log and all scripts are released under Apache-2.0. The sEMG and MEG data show that one format and one code path work across devices; they do not show that the commit rule works across decoders.

Figure 1 summarises the result. On the one intracortical participant, the layer turned frequent wrong commits into rare ones. On composed sEMG sentences it committed few sentences, and on MEG it never committed. Its speed gains over the same policy without commits were small (about 1% and 3%) and depend on the user model, and on the published decoder always committing was faster, so our claim is about safety.

2 Related work

Brain-to-text decoders. MEG decoding of typed sentences reached a CER of 0.29 with a keystroke-locked model [6]; a second version decodes without key times and adds an LLM stage [7]. Intracortical decoders reach high accuracy for handwriting, speech and typing [1, 2, 4, 5, 28], with public phoneme logits [29, 30]; sEMG text entry exists for the wrist and forearm [8–10]. Most use CTC [31] with beam search and an LM [32]. In the intracortical study whose data we use, a sentence was finalised after 6 s without speech or by an eye-gaze button, then read aloud, and the participant rated it afterwards [2]. There was no confirmation before text-to-speech, which corresponds to our COMMIT. Other speech neuroprostheses make the same decision implicitly [3, 18, 33].

Error handling in BCIs and speech interfaces. P300 and RSVP spellers commit a character when its posterior passes a threshold [34–36], often with an LM prior [37–39]. Error-related potentials allow automatic error detection [40–43]; asynchronous BCIs detect a "no-control" state [44–46]; myoelectric control rejects uncertain decisions [47]; adaptive decoders change the decoder over days [48–52]. Kim et al. accept or abstain on single EEG speech classifications with calibrated ensembles [24], and Huang et al. give calibrated alternatives [25]; neither bounds the rate of wrong sentences sent. Speech recognition has a long line of word and utterance

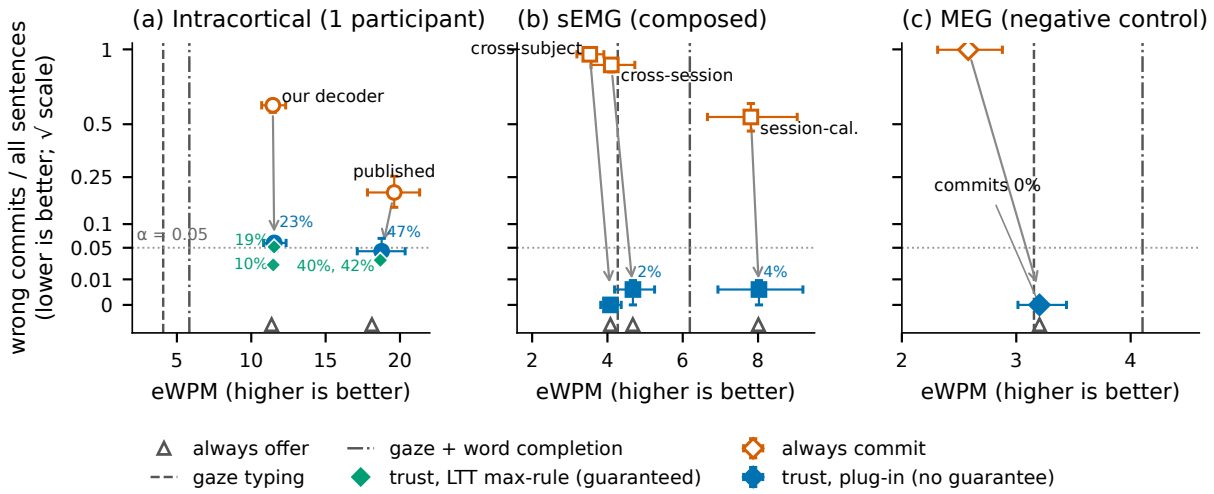


Figure 1: The trust layer turns frequent wrong commits into rare ones; on MEG it does so by never committing. Wrong commits per sentence attempted (signed square-root axis) against effective words per minute (eWPM) of a simulated user. Open markers: always commit; blue: plug-in trust layer (full fit, no guarantee); green diamonds: the guaranteed rule (fit-A pipeline, guard, clean calibration, LTT max rule; fixed splits A and B); triangles: always offer (no wrong commits; drawn just below 0). Labels give the commit rate. Dashed line: gaze typing; dash-dotted: gaze typing with word completion; dotted: $\alpha = 0.05$. Error bars: 95% cluster-bootstrap CIs over days (a), participants (b) and sentences × participants (c), conditional on the fitted pipeline. (a) Intracortical speech, one participant, 816 test trials. (b) sEMG, 551 composed sentences, 19 participants; the guaranteed rule gives the same rates as the plug-in rule (not drawn). (c) MEG typing, 637 instances of 17 sentences; the layer equals always offer.

confidence measures [53], and spoken dialogue systems accept, confirm or reject an utterance by cost-based thresholds [54]. Our commit / offer / abstain decision is this accept / confirm / reject decision with a finite-sample bound.

Calibration, selection and risk control. Temperature scaling fixes over-confidence [19] but degrades under shift [55]. Selective classification trades coverage for risk; selection with guaranteed risk (SGR) bounds the error among accepted items [20]. CRC bounds the expected value of a monotone loss [22, 56], LTT bounds a risk with high probability [23], weighted variants give approximate bounds under shift [57–59], and online methods give long-run guarantees for any sequence [60–63]. Conformal selection controls the false discovery rate among selected items [64], and conformal alignment applies it to decide when to trust foundation-model outputs [65]; this expected error among selected items is the natural comparator for our error among commits. Conformal methods were applied to speech recognition [26, 66] and to LM outputs and abstention [67–69]. We use these methods as published; our contribution is their use and evaluation as a commit rule.

Metrics. BCI utility metrics [70] and text-entry metrics [71] charge for corrections, unlike the information transfer rate [72]; our eWPM is close to the utility metric. LM prediction speeds up gaze typing [27, 73], but scanning prediction lists has a cost [74].

3 Problem setting and stream interface

Items, streams and actions. An *item* is one attempt to enter one sentence: a *trial* of the intracortical data, an *instance* (one participant typing one sentence) of the MEG data, or a composed *sentence* of the sEMG data. For item i, the decoder emits a stream of F_i frames of

log-probabilities $\log p_t(v)$ over V tokens. In our open format, NeuralTextStream v0.1, a stream holds this $F \times V$ array, the frame times, the typed vocabulary and the CTC blank index. A producer must emit the decoder's own, unmodified distribution, and ground-truth text must not appear in the stream. One small adapter per device maps decoder output into the stream; every later step is the same code for all devices, with parameters re-fitted per device (format, conformance suite, adapters and audit log: Section S18).

An LM search returns $K = 10$ ranked candidate sentences with scores, from which we compute $\hat{p}_{\text{top1}}$, the calibrated probability of the top candidate. The system then takes one action: COMMIT the top candidate (sent without further input); OFFER up to k candidates for the user to pick; or ABSTAIN, after which the user types with a fallback keyboard. Let $c_i \in \{0, 1\}$ indicate that the top candidate differs from the intended sentence (exact match after normalising case and punctuation).

The two error quantities. Over the n_{test} test items,

$$\text{wrong-commit rate} = \#\{\text{wrong commits}\}/n_{\text{test}} \approx \Pr(\text{commit} \wedge \text{wrong}), \tag{1}$$

$$\text{error among commits (EAC)} = \#\{\text{wrong commits}\}/\#\{\text{commits}\} \approx \Pr(\text{wrong} \mid \text{commit}). \tag{2}$$

The first is the *marginal* risk: how often wrong sentences reach the recipient per sentence attempted. It is monotone in the commit threshold, so CRC and LTT can bound it (Section 4.2). Committing rarely makes it small, so we always report it with the *commit rate*. The second is the *selective* risk that a user sees in committed text (Section 4.3). $\alpha = 0.05$ is the target wrong-commit rate, β the target error among commits and $\delta = 0.1$ the failure probability of the high-probability guarantees (notation: Table S1).

4 The trust layer

The trust layer reads one stream per item and returns one action. It never reads raw signals or decoder weights. Algorithm 1 (Appendix A) gives the decision for one item, and Algorithm S1 the fitting and calibration. All parameters are fitted per device on that device's fit data (fitted values: Table S5; policy grid: Section S6).

4.1 Calibration, candidates, guard and policy

Decoder log-probabilities are often over-confident. A frame temperature τ_f rescales every frame, $\tilde{p}_t(v) \propto p_t(v)^{1/\tau_f}$; it minimises the cross-entropy of the aligned true tokens at the greedy CTC peaks. It is applied *before* the LM search, so it changes the decoded text as well as the confidence. A sentence temperature τ_s on the scores of the K-best list gives $\hat{p}_{\text{top1}} = \text{softmax}(\text{scores}/\tau_s)_1$; it minimises the binary negative log-likelihood of "the top candidate is right" [19].

Candidates come from a CTC prefix beam search with a character 6-gram LM trained on Tatoeba sentences [31, 32]; for intracortical speech, a phoneme beam search is followed by a CMUdict lexicon, a word unigram and the same character LM. Each character of the top candidate gets an acoustic confidence from the calibrated frame posteriors. A *guard* scores how much of the top candidate the LM invented: the larger of the share of characters with acoustic confidence below 0.5 (*low evidence*) and one minus the posterior-weighted word agreement between the top candidate and the top five. The policy takes, in this order: ABSTAIN if the guard risk is at least r_{abst}; COMMIT if $\hat{p}_{\text{top1}} \geq \lambda$ and the guard risk is at most $r_{\max}$; OFFER a split-conformal sentence set if it has between 1 and $k_{\max}$ candidates; otherwise a fallback. Its thresholds maximise eWPM (Section 5) on fit data, subject to a fit-data wrong-commit rate of at most α. This *plug-in* fit has no guarantee on new data. A *plain threshold* is the same policy without the guard.

The guard was part of the configuration fixed before the evaluation. On real intracortical data it did not lower the wrong-commit rate. On no-information streams it is the only part that stops commits (Section 6.6), so we keep it as an input-validity check.

Gate on a published sentence: a two-decoder agreement signal. For the published intracortical decoder we have the stream and the published LM-rescored sentence, but not the decoder's scores, LM or weights. We put the published sentence on top of the K-best list of our own phoneme-to-word decoder, which runs on the same stream, with a score equal to our best score plus a bonus of 2.0, and give each published word the acoustic confidence of the aligned word in our top candidate. The confidence of a published sentence is therefore a measure of agreement between the two decoders; it is not a stream-only confidence of the published decoder. The published sentence alone gives the layer no signal, and it then never commits (Section S11). The bonus of 2.0 was set by hand when the gate was first written (commit `3fb8154`) and was never tuned; bonuses from 0.5 to 4 change the main rows little (Table S11).

4.2 Finite-sample control of the wrong-commit rate

The commit score s_i is $\hat{p}_{\text{top1}}$ if the other policy rules allow a commit, and $-\infty$ otherwise. The layer commits when $s_i \geq \lambda$, with loss $L_i(\lambda) = c_i \, \mathbf{1}[s_i \geq \lambda]$. L_i is non-increasing in λ, and "never commit" ($\lambda = +\infty$) has loss 0. On n calibration items with known truth and empirical risk $\hat{R}_n(\lambda)$:

- **CRC** [22] chooses the smallest λ among the calibration scores and $+\infty$ with $\frac{n}{n+1}\hat{R}_n(\lambda) + \frac{1}{n+1} \leq \alpha$. If the calibration items and a new item are exchangeable, $\mathbb{E}[L_{n+1}(\hat{\lambda}_{\text{CRC}})] \leq \alpha$ over the calibration draw and the new item. This bounds a mean; a single calibration set can exceed α. CRC never commits for $n < 19$. Because $\hat{\lambda}_{\text{CRC}}$ is a minimum over a finite set of scores, the bound holds for the rule "commit if $s \geq \hat{\lambda}$" as implemented, and test items below the lowest eligible calibration score are never committed.
- **LTT** [23] tests $H_0(\lambda)$: $R(\lambda) > \alpha$ along one fixed, pre-specified sequence from $\lambda = 1.00$ down to 0.00 in steps of 0.01, with the exact binomial p-value $\Pr(\text{Bin}(n, \alpha) \leq n\hat{R}_n(\lambda))$, and stops at the first non-rejection; $\hat{\lambda}_{\text{LTT}}$ is the last rejected λ. Fixed-sequence testing needs no multiplicity correction. If the calibration items are i.i.d. from a distribution P, then $\Pr(R_P(\hat{\lambda}_{\text{LTT}}) \leq \alpha) \geq 1 - \delta$. LTT cannot commit below $n = 45$.
- **Max rule.** The deployed threshold is $\max(\lambda_{\text{tuned}}, \hat{\lambda})$, where λ_{tuned} is the plug-in threshold. A larger threshold has a pointwise smaller loss, so both guarantees carry over. "Pure" CRC or LTT drop λ_{tuned}.

Scope of the guarantee. P is the distribution that the calibration items come from. In our data it is a mixture of days (intracortical) or participants (sEMG). LTT therefore bounds the risk averaged over that mixture. It does not bound the risk on each day, and a user meets one session at a time (Section 6.3). Per-session control needs per-session (Mondrian) calibration, with at least 45 calibration items per session for LTT, or a calibration block at the start of each session.

Primary rule. Our primary rule is LTT with the max rule, calibrated on clean data. The theory gives an a-priori reason (a high-probability bound instead of a mean), but we chose it in revision 3, after we saw the random re-splits of Section 6.3; it is exploratory like every other result here. CRC with the max rule is reported next to it.

4.3 Controllers for the error among commits

The rules above do not bound the error among commits. Its selective risk $\Pr(\text{wrong} \mid s \geq \lambda)$ is not monotone in λ, so a max rule does not keep a selective bound. Our reference selective controller is *LTT-sel*: on the fixed grid $\lambda = 0.05, 0.10, \ldots, 1.00$ it tests $H_0(\lambda)$: $\Pr(\text{wrong} \mid s \geq \lambda) > \beta$ with the exact binomial p-value given the n_λ calibration commits, with a Bonferroni correction over the 20 points, and takes the smallest rejected λ. *Joint LTT* tests both targets at once: $H_0(\lambda)$: $R(\lambda) > \alpha$ or $\Pr(\text{wrong} \mid s \geq \lambda) > \beta$, with $p(\lambda) = \max(p_{\text{marg}}, p_{\text{sel}})$, the same grid and Bonferroni. With probability at least $1 - \delta$, both risks at the chosen λ are within target. For comparison we report SGR [20], whose guarantee is the one stated in their Theorem 3.2; its bisection chooses the tested thresholds from the data. A selective bound conditions on committing, so it needs many calibration commits.

Separate data. Both guarantees need the parts that define s_i to be fitted on data disjoint from the calibration items. The protocol ensures this, with four exceptions (Section S19): on Brain-to-Text '25 the word-decoder grid was tuned on fit trials that later fall into calibration or test sets (fixed splits and random re-splits); the MEG checkpoint was selected on validation data that include the calibration sentences; and sEMG session-calibrated uses two cross-fitted decoders. An online controller that re-tunes the threshold from user feedback is a safeguard with a long-run guarantee (Section S15).

5 Evaluation protocol

5.1 Outcomes, user model and baselines

The primary safety outcome is the wrong-commit rate, always reported with the commit rate and the error among commits. Speed is secondary. We simulate a user who reads the output, accepts or corrects a commit, picks from an offer, or types the sentence with a fallback keyboard. For item i with intended sentence y_i, the simulation returns the total time t_i and the final text, and

$$\text{eWPM} = 60 \cdot \frac{\sum_i \mathbf{1}[\text{final text}_i = y_i] \ \text{words}(y_i)}{\sum_i t_i}. \tag{3}$$

Like the BCI utility metric [70], eWPM charges the time needed to fix errors. A *silent error* is a wrong final text that the user did not correct; we report its rate separately.

The main user types with a gaze keyboard at 20 characters per minute and scans and selects one of $k = 3$ word alternatives in 4.5 s. The typing speed was measured in one eye-gaze user with ALS, and the 4.5 s are close to the 4.7 s measured in the same study [27]; all other costs are assumptions (Table 5). After a COMMIT, the user misses each wrong word with probability 0.05; after an OFFER, the main user sees every error. This asymmetry favours offers and is not measured. The neural input runs at 159.3 (intracortical), 75.8 (sEMG) and 152 (MEG) characters per minute. The intracortical rate is the number of characters of the cue sentences divided by the sum of the `speech_duration_s` field of the data file. The file does not say whether this field includes the 6 s finalisation pause of the original interface, and we add no time for that pause, which favours the neural input. Because only two costs are measured, we refit every policy under eight user-model variants (Section S4).

Baselines: always commit the top candidate; always offer and never commit; the *same fitted policy without commits* (the same fit-A pipeline and policy with $\lambda = +\infty$, so every item takes the policy's non-commit path), which is the paired comparator for the speed of a commit rule; the plain threshold; gaze typing alone (error-free, which favours it); and gaze typing with *word completion*, where the keyboard shows the three most frequent words with the typed prefix (a context-free unigram).

5.2 Datasets

Intracortical speech (Brain-to-Text '25). The public copy-task file holds the online phoneme CTC logits, cue sentences and published LM-rescored sentences of participant T15 of the BrainGate2 clinical trial [2, 30]. We use the 1,628 trials with the 125,000-word vocabulary (14 post-implant days, 30–208 trials each). The file has no time stamps, so the file order within each day is the trial order. We evaluate our phoneme-to-word decoder (test WER 0.149) and the published sentence (WER 0.043), gated as in Section 4.1. A trial is "wrong" if the top sentence differs from the cue; the file has no per-trial participant ratings, so we cannot check this label against the participant's own judgement.

Forearm sEMG and MEG typing. The Toronto sEMG data hold isolated key presses from 19 participants over two days [9, 75]. We compose held-out Tatoeba sentences from single presses with oracle timing and no inter-key context (570 fit, 551 test sentences), under three decoder conditions: *session-calibrated*, *cross-session* and *cross-subject*. In the SpanishBCBL MEG data, healthy participants typed memorised Spanish sentences [6, 76]; our asynchronous decoder gets no key times at inference, and the test set has 637 instances of 17 sentences (Section S17). We use sEMG and MEG to test the stream format and the safety of the layer on weak decoders, not to test the commit rule across decoders.

5.3 Calibration data and deployment tests

The *deployment protocol* calibrates on data that played no other role: a fit-A part fits the temperatures, conformal sets, guard and policy, and a disjoint calibration part only calibrates.

Fixed splits. On Brain-to-Text '25 the pipeline is fitted on the first half of each day (812 trials) and tested on the second half (816). Each day's fit trials are split into fit-A (411 trials) and clean calibration (401), by alternating trials (split A) or by taking the earlier half as fit-A (split B). On sEMG, 142 fit sentences fit the policy and 143 calibrate; on MEG, 6 validation sentences fit and 5 calibrate.

Time-ordered (forward-chaining) tests. A real deployment calibrates on earlier data and runs on later data. On Brain-to-Text '25 we therefore fit, calibrate and test in day order: (V1) fit on days $1..k-2$, calibrate on day $k-1$, test on day k; (V4) fit on days $1..k-3$, calibrate on days $k-2$ and $k-1$; (V2) fit once on days 1–3, calibrate on all earlier non-fit days; (V3, session start) fit on days $1..k-1$, calibrate on the first half of day k and test on its second half. These give 10–13 test days per design. sEMG has no session order (sentences are composed from presses drawn at random), so it has no time-ordered test.

Random re-splits. We also repeat the protocol 50 times with random assignments. On Brain-to-Text '25, each re-split permutes every day's trials and splits them day-stratified into fit-A (1/4), clean calibration (1/4) and test (1/2); on sEMG, each participant's fit sentences are re-split into fit-A, policy and calibration parts, with the fixed 551 test sentences, so the sEMG spread is fit and calibration variation only. To cache decodes, the fitted frame temperature was rounded to a step of 0.02 (Brain-to-Text '25) or 0.01 (sEMG); the largest relative change was 0.27% (Brain-to-Text '25) and 0.38% (sEMG). Calibration and test items are exchangeable by construction, so these re-splits are a Monte-Carlo check of the theorems on one pool of data, not a deployment test. Their spread is split-to-split variation, not a CI for new data. We also report the *population* exceedance: we draw calibration sets with replacement from a split's pool, so that the pool is the exact population, and evaluate the chosen threshold on the whole pool (Section S10).

5.4 Controls and statistics

Signal-destroying controls keep the stream format but remove the neural information (frame-shuffled and near-uniform streams, zero-input output, a permuted-label decoder); a stream-level layer should not commit these. The *intent-mismatch* control is the real stream of a different sentence by the same participant, which a stream-level layer cannot detect.

All bootstrap CIs are 95% percentile intervals from one bootstrap ($B = 2000$, seed 20261112) at the primary unit: days on Brain-to-Text '25 (14 clusters), participants on sEMG (19), sentences and participants at once on MEG. They are conditional on the fitted pipeline and threshold. Because 14 clusters are few, we add exact day-level sign tests for key paired differences. Single rates and shares of splits or days get exact Clopper–Pearson (CP) 95% intervals; rates with no events get one-sided 95% CP upper bounds (items treated as independent, which is optimistic under clustering). The primary endpoints were named in revision 1, after the first evaluation runs; the deployment protocol, the re-splits, the time-ordered tests, the selective controllers and the choice of the primary rule came in later revisions. All inference is therefore exploratory. All decoder settings, user-model values, the policy grid, α, β and δ were fixed before the runs that use them.

6 Results

6.1 The decoders use the neural signal

On Brain-to-Text '25, our phoneme-to-word decoder reached WER 0.149 with the right sentence first for 39% of test trials, and the published decoder WER 0.043 with 81%. On sEMG, balanced key accuracies (0.367, 0.240, 0.226) were well above chance (0.037), and after the LM the right sentence was first for 46%, 12% and 4% of composed sentences. Our asynchronous MEG decoder reached a test CER of 0.475 (shuffled MEG 0.760, zero input 0.925; three seeds and controls: Tables S3 and S4), but its top-1 sentence was right for only 2 of 637 instances.

6.2 Time-ordered deployment: the guaranteed rule held across days

Table 1 and Figure 2a give the time-ordered tests on Brain-to-Text '25, where calibration always came before the test data. Under the LTT max rule, the pooled wrong-commit rate was 0.010–0.027 in every design, on both decoders. It was above α on 0–2 of 10–13 test days, and no day had a CP lower bound above α; the highest single day was 0.060. With one previous day as calibration (V1), it committed 11% (ours) and 32% (published) of trials, and 20% and 5% of these commits were wrong. The plug-in rule was above α on up to 6 of 10 days, and in three of the four designs one day had a CP lower bound above α (ours). CRC with the max rule lay between the two. With the guard off (plain threshold), LTT max was above α on 0 of 12 days in V1 (pooled 0.018 ours, 0.010 published) (Table S7). Twelve test days can neither confirm nor exclude control at level δ per day: 2 of 12 days has a CP interval of 0.02–0.48.

6.3 Random re-splits and the scope of the guarantee

Over 50 random re-splits (Table 2), the LTT max rule gave mean wrong-commit rates of 0.030 (our decoder) and 0.031 (published gate). It exceeded α on the test set in $4/50 = 0.08$ [0.02, 0.19] and $1/50 = 0.02$ [0.001, 0.11] of re-splits. Fifty re-splits cannot show that this share is at most $\delta = 0.1$, and the test-set share includes test-sampling noise. The exact population exceedance, which LTT bounds, was 0.026 and 0.014. CRC met its own target, the mean (0.046 and 0.049), and, as designed, many single re-splits were above α (34% and 42%). LTT max was never above α in a re-split where CRC max was not; with 50 re-splits this difference was significant only on the published gate (exact McNemar $p = 0.25$ and 0.031; with 200 re-splits $p < 10^{-6}$ on both, Section S10). With the guard off, LTT max exceeded α in 0 of 50 re-splits on both decoders. On

Table 1: Time-ordered (forward-chaining) deployment on Brain-to-Text '25 (guard on). Calibration data always come before the test data. Days $> \alpha$: test days with a wrong-commit rate above $\alpha = 0.05$ (none had a CP lower bound above α under LTT max). Pooled: wrong-commit rate over all test trials of the design, with exact CP 95% CI. Commit, EAC: pooled. Joint LTT: commit rate with $\beta = 0.10$. Guard off and all rules: Table S7.

		Plug-in	CRC max	LTT max (primary)				Joint LTT
Decoder	**Design (test days)**	**Days $> \alpha$**	**Days $> \alpha$**	**Days $> \alpha$**	**Pooled [CP95]**	**Commit**	**EAC**	**Commit**
Ours	V1: previous day (12)	5	4	**2**	0.022 [0.015, 0.031]	0.11	0.20	0
	V4: two previous days (11)	5	4	**0**	0.027 [0.019, 0.037]	0.14	0.20	0
	V2: all earlier days (10)	6	4	**1**	0.027 [0.019, 0.038]	0.15	0.18	0
	V3: session start (13)	5	3	**2**	0.015 [0.008, 0.026]	0.09	0.17	0
Published gate	V1: previous day (12)	1	1	**0**	0.016 [0.010, 0.024]	0.32	0.05	0
	V4: two previous days (11)	0	0	**0**	0.020 [0.013, 0.029]	0.35	0.06	0.04
	V2: all earlier days (10)	0	0	**0**	0.023 [0.016, 0.033]	0.29	0.08	0.06
	V3: session start (13)	1	1	**1**	0.010 [0.004, 0.020]	0.19	0.05	0

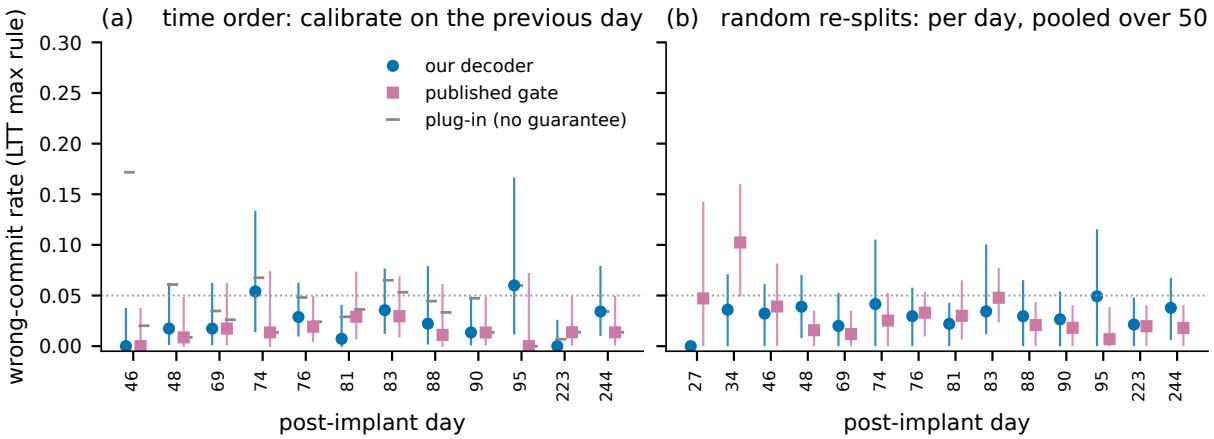


Figure 2: The guaranteed rule across days (Brain-to-Text '25, guard on, LTT max rule; dotted line: $\alpha = 0.05$). (a) Time order (V1: fit on days $1..k-2$, calibrate on day $k-1$, test on day k): wrong-commit rate per test day with exact CP 95% CIs; small grey markers: plug-in rule. (b) Random re-splits: wrong-commit rate per day pooled over 50 re-splits; bars: 5th–95th percentile over re-splits. The bound covers the mix of days; on the published gate, day 34 ran at about 2α.

sEMG, LTT max committed 9% and 2% of sentences (session-calibrated, cross-session; mean wrong-commit 0.011 and 0.006) and never exceeded α; on cross-session, 34% of re-splits never committed, and the cross-subject decoder never committed.

Per day, the risk was not controlled (Figure 2b, Table S8). On the published gate, day 34 (about 50 test trials per re-split) had a pooled wrong-commit rate of 0.102 under LTT max, and it was above α in 94% of re-splits; in fixed splits A and B it had 8 wrong commits in 50 trials (0.16, CP95 [0.07, 0.29]). The other days had 0.007–0.048, and our decoder 0.000–0.049 on every day. LTT bounds the risk over the mix of days that the calibration set comes from, not each day. Per-day (Mondrian) LTT with a quarter of each day as calibration was feasible only on the one day with at least 45 calibration items, and committed nothing on the other 13 days.

The guaranteed rule was faster than the same fitted policy without commits in every re-split: by +0.114 eWPM (Monte-Carlo SE 0.005) on our decoder and +0.510 (SE 0.007) on the published gate, about 1% and 3%. On the fixed test set, the paired gains were +0.058 [0.034, 0.090] and +0.125 [0.077, 0.188] (ours, splits A and B) and +0.537 [0.439, 0.636] and +0.567 [0.456, 0.694] (published).

6.4 Fixed test sets; sEMG and MEG as format and safety checks

On the fixed test set (Table 3), always committing sent wrong text for 0.610 of trials on our decoder and 0.194 on the published sentence, because one wrong word makes a sentence wrong.

Table 2: Random re-splits (50; exchangeable by construction, so a check of the theorems on one data pool, not a deployment test). Wrong-commit: mean test rate (5th–95th percentile over re-splits); $> \alpha$: share of re-splits with a test rate above α, with exact CP 95% CI; Pop.: share of with-replacement calibration draws whose exact population risk exceeds α (Section S10); Commit: mean; EAC: pooled; Gain: mean paired eWPM gain over the same fitted policy without commits. The spread is split-to-split variation in one data set, not a CI for new data. sEMG rows: Table S9.

Rule	Wrong-commit	$> \alpha$ [CP95]	Pop.	Commit	EAC	Gain
Brain-to-Text '25, our decoder						
plug-in (fit-A)	0.044 (0.021–0.072)	0.34 [0.21, 0.49]	—	0.19	0.24	+0.132
CRC, pure	0.046 (0.026–0.068)	0.34 [0.21, 0.49]	0.39	0.19	0.24	—
CRC, max rule	0.037 (0.021–0.059)	0.14 [0.06, 0.27]	0.17	0.17	0.22	+0.125
LTT, max rule	**0.030** (0.016–0.052)	**0.08** [0.02, 0.19]	0.026	0.15	0.21	+0.114
LTT, max rule, guard off	0.031 (0.013–0.043)	0.00 [0, 0.07]	—	0.14	0.21	+0.106
Brain-to-Text '25, published gate						
plug-in (fit-A)	0.039 (0.021–0.060)	0.22 [0.12, 0.36]	—	0.43	0.09	+0.564
CRC, pure	0.049 (0.029–0.075)	0.42 [0.28, 0.57]	0.40	0.48	0.10	—
CRC, max rule	0.036 (0.021–0.055)	0.14 [0.06, 0.27]	0.074	0.42	0.09	+0.546
LTT, max rule	**0.031** (0.020–0.047)	**0.02** [0.001, 0.11]	0.014	0.39	0.08	+0.510
LTT, max rule, guard off	0.024 (0.018–0.031)	0.00 [0, 0.07]	—	0.36	0.07	+0.471

The guaranteed rule gave 0.025 and 0.052 (splits A and B) on our decoder and 0.029 and 0.031 on the published gate. The plug-in layer fitted on all fit trials gave 0.059 [0.045, 0.070] on our decoder, above α. The difference to the fit-A rows comes from the policy grid fitted on fit-A, not from the risk control: in split A, CRC and LTT were not binding (Section S7). On sEMG, the layer cut the wrong-commit rate from 0.541, 0.882 and 0.962 to 0.004, 0.004 and 0, committing 21, 10 and 0 of 551 sentences; the guaranteed rule gave the same rows. On MEG the layer committed nothing (0 of 637, one-sided 95% upper bound 0.0047) and equals always offering; no guarantee for new sentences is possible on these data, because a sentence-level LTT needs at least 45 calibration sentences and the data set has 17 test sentences (Section S17). Only the session-calibrated sEMG decoder clearly beat gaze typing [+2.71, +4.91]; on the 256 sEMG test sentences without a near variant in the LM text, speeds were lower and gaze typing beat the cross-subject decoder (Section S16).

6.5 A guaranteed error among commits of 10% almost never commits

The marginal rules let a weak decoder commit text that is often wrong: 21% of the guaranteed rule's commits on our decoder (Table 2). Joint LTT bounds both risks (Table 4). With $\beta = 0.10$ it committed nothing on our decoder, sEMG and MEG, nothing in either fixed split, and nothing in any time-ordered design with one day of calibration. Over random re-splits of the published gate, it committed 7.2% of sentences on average (none in 76% of re-splits), with an error among commits of 0.059. With $\beta = 0.20$, it committed 32% of published sentences with an error among commits of 0.062, and 0.32 in fixed split A (error among commits 0.057 [0.032, 0.093]); our decoder still committed 0.1%. LTT-sel, which bounds only the selective risk, committed 56% of published sentences at $\beta = 0.20$, but its marginal rate (0.074) exceeded α in 76% of re-splits. The limit is the calibration size: an exact bound on the error among commits needs many calibration commits.

6.6 Controls: without the guard, the layer commits no-information streams

Always committing wrote a sentence for every control stream. With the guard, the guaranteed rule (split A) committed none of the 300 frame-shuffled and none of the 300 uniform streams on either Brain-to-Text '25 decoder (one-sided 95% upper bound 0.0099), while it committed

Table 3: Safety and speed on the fixed test sets. Plug-in: full fit, no guarantee. Guaranteed rule (the v4 frozen configuration): fit-A pipeline, guard, clean calibration, LTT max rule (B2T '25: splits A and B, $n_{\text{cal}} = 401$; sEMG: 143). Guard off: the same with the plain threshold. EAC: error among commits (wrong/commits in brackets; count only if fewer than 30 commits). Silent: uncorrected wrong final texts per item. 95% cluster-bootstrap CIs over days (B2T '25), participants (sEMG) or sentences × participants (MEG), conditional on the fitted pipeline and threshold. Zero rates: one-sided 95% upper bounds 0.0037 (816 items), 0.0054 (551) and 0.0047 (637). All policies: Table S6.

Device	Policy	eWPM [CI]	Wrong-commit [CI]	Commit	EAC	Silent
B2T '25, our decoder	always commit	11.45 [10.70, 12.31]	0.610 [0.568, 0.647]	1.00	0.61	0.044
	always offer	11.37 [10.67, 12.17]	0	0	—	0
	plug-in	11.55 [10.83, 12.34]	0.059 [0.045, 0.070]	0.23	0.25 (48/189)	0.003
	guaranteed rule, A	11.48 [10.79, 12.29]	**0.025** [0.015, 0.035]	0.10	0.24 (20/82)	0.001
	guaranteed rule, B	11.54 [10.83, 12.34]	**0.052** [0.043, 0.059]	0.19	0.27 (42/157)	0.003
	guard off, A	11.55 [10.83, 12.39]	0.040 [0.030, 0.051]	0.18	0.22 (33/149)	0
	guard off, B	11.53 [10.82, 12.37]	0.029 [0.021, 0.038]	0.15	0.20 (24/120)	0
B2T '25, published	always commit	19.61 [17.81, 21.32]	0.194 [0.146, 0.253]	1.00	0.19	0.025
	always offer	18.11 [16.59, 19.57]	0	0	—	0
	plug-in	18.77 [17.13, 20.35]	0.044 [0.027, 0.068]	0.47	0.09 (36/386)	0.004
	guaranteed rule, A	18.65 [17.04, 20.20]	**0.029** [0.013, 0.050]	0.40	0.07 (24/329)	0.001
	guaranteed rule, B	18.68 [17.06, 20.23]	**0.031** [0.014, 0.052]	0.42	0.07 (25/339)	0.003
	guard off, A	18.63 [17.03, 20.18]	0.027 [0.016, 0.040]	0.39	0.07 (22/319)	0.001
	guard off, B	18.63 [17.02, 20.18]	0.025 [0.014, 0.038]	0.39	0.06 (20/316)	0.001
B2T '25	gaze typing	4.08 [4.05, 4.12]	0	—	—	0
sEMG session-cal.	always commit	7.81 [6.66, 9.04]	0.541 [0.463, 0.621]	1.00	0.54	0.060
	guaranteed rule	7.98 [6.95, 9.11]	**0.004** [0, 0.009]	0.04	2/21	0
sEMG cross-session	always commit	4.10 [3.56, 4.73]	0.882 [0.831, 0.928]	1.00	0.88	0.149
	guaranteed rule	4.68 [4.19, 5.25]	**0.004** [0, 0.009]	0.02	2/10	0
sEMG cross-subject	always commit	3.53 [3.19, 3.90]	0.962 [0.942, 0.981]	1.00	0.96	0.167
	guaranteed rule	4.08 [3.82, 4.37]	**0**	0	—	0
sEMG	gaze typing	4.27 [4.22, 4.32]	0	—	—	0
MEG (asynchronous)	always commit	2.58 [2.31, 2.88]	0.997 [0.987, 1.000]	1.00	1.00	0.235
	trust (= always offer)	3.20 [3.01, 3.44]	0	0	—	0
MEG	gaze typing	3.15 [3.02, 3.31]	0	—	—	0

29 (ours) and 115 (published) of the 300 matched real trials; the full-fit layer also committed none, and so did sEMG (0 of 200 sentences per control; bound 0.015) (Figure S6). Without the guard, our decoder committed 198 of 300 uniform streams under the plain threshold with the LTT max rule (all 198 commits wrong) and 3 of 300 frame-shuffled streams. A uniform stream gives flat phoneme posteriors; the word decoder then returns a fluent LM sentence with a high calibrated $\hat{p}_{\text{top1}}$, and the calibration data, which contain only real streams, cannot detect this. The published gate, sEMG and MEG committed no control stream with or without the guard. On real data, in contrast, the guard did not lower wrong commits. Without it, the full-fit wrong-commit rate on our decoder fell from 0.059 to 0.036 (paired $[-0.035, -0.011]$; sign test over days $p = 0.021$). In the re-splits, the time-ordered tests and the fixed splits, the rates without the guard were lower on the published gate in 6 of 7 comparisons and mixed on our decoder (lower in 2, similar in 2, higher in 3; fixed split A: 0.040 against 0.025) (Tables 2, 3 and S7). A commit rule therefore needs an input-validity check that real calibration data do not test; our guard is one such check, and a better one is open. Given the real stream of another sentence (intent mismatch), the layer committed at the matched rate on our decoder: nothing in the stream reveals the intent. The published gate committed none of these 300 streams (Table S19).

6.7 Speed depends on the user model; wrong-commit rates do not

Figure 3 refits every policy (plug-in layer) under eight user-model variants. The layer was at least as fast as always offering in every variant and condition. Always committing was faster than the layer on the published gate in all 9 variants (by 0.46–1.47 eWPM, every CI below 0) and on our decoder in 5 of 9 (CI below 0 in 1, above 0 in 3). Gaze typing with word completion had a higher point value than the layer in every variant on MEG and on the two weaker sEMG decoders. The

Table 4: Controlling the error among commits (B2T '25 published gate, 50 random re-splits, guard on, $\delta = 0.1$; test set). Commit: mean (share of re-splits with no commit); EAC: pooled error among commits; $> \beta$: committing re-splits with EAC above β; Marginal: mean wrong-commit rate ($> \alpha$: share of re-splits above α). Joint LTT: both targets, fixed grid, Bonferroni (valid for both). LTT-sel: selective target only. SGR: bisection [20]. Our decoder: joint LTT commits nothing at $\beta = 0.10$ and 0.1% at $\beta = 0.20$; sEMG and MEG: no commits at any β.

β	Rule	Commit (none)	EAC	$> \beta$	Marginal	$> \alpha$	eWPM
0.10	**joint LTT**	0.072 (0.76)	0.059	0 of 12	0.004	0	18.07
	LTT-sel	0.072 (0.76)	0.059	0 of 12	0.004	0	18.07
	SGR	0.112 (0.64)	0.069	0 of 18	0.008	0	18.12
0.20	**joint LTT**	0.318 (0.02)	0.062	0 of 49	0.020	0	18.39
	LTT-sel	0.564 (0)	0.131	0 of 50	0.074	0.76	18.72
	SGR	0.617 (0)	0.142	0 of 50	0.088	0.92	18.79
—	LTT max rule (marginal only)	0.392 (0)	0.078	—	0.031	0.02	18.49
—	no commits	0 (1)	—	—	0	0	17.98

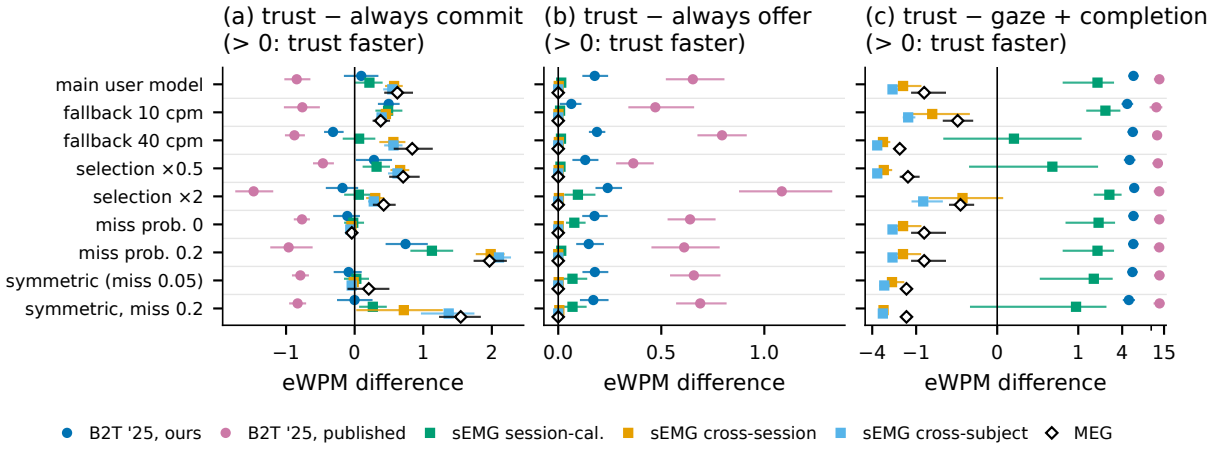


Figure 3: User-model sensitivity (plug-in layer, full fit; not the guaranteed rule). Paired eWPM difference between the trust layer and (a) always committing, (b) always offering and (c) gaze typing with word completion (symmetric-log axis), all policies refitted under each variant (rows); values above 0 mean that the layer is faster. 95% cluster-bootstrap CIs at the primary unit.

wrong-commit rate of the refitted layer was at most 0.059 in all 54 condition–variant pairs.

6.8 Calibration and the other components

Sentence ECE fell from 0.448 [0.419, 0.475] to 0.067 [0.065, 0.104] on our intracortical decoder and from 0.102 [0.058, 0.161] to 0.058 [0.044, 0.090] on the published gate (percentile CIs; binned ECE is biased upwards, so an interval can lie beside its point value). The calibrated $\hat{p}_{\text{top1}}$ predicted a wrong top sentence with AUROC 0.797 [0.773, 0.819] (ours), 0.710 [0.678, 0.744] (published) and 0.828, 0.755 and 0.674 on sEMG (Section S12); the MEG AUROC is not informative (2 right sentences). The frame temperature raised the ECE of per-frame phoneme peaks (0.090 to 0.130) although it minimises their cross-entropy; we did not resolve why, but the two criteria differ (cross-entropy at aligned peaks with blank targets for inserted peaks, ECE of the arg-max confidence). Without the sentence temperature, wrong commits rose on 4 of 6 conditions (Table S13). Calibrating pure CRC on data that had also fitted the pipeline raised the test rate on our decoder by +0.009 (day-jackknife 95% CI [+0.005, +0.013]; Holm-adjusted $p = 0.004$ over 12 comparisons), conditional on this data pool; no other condition showed an effect after Holm, and with 200 re-splits the effect was +0.003 (Table S14). As the threshold varies, the published decoder can commit about half of its sentences at the 0.05 marginal level

and our decoder about a fifth (Figure S3).

7 Discussion

The layer is a safety mechanism, tested on one person. All non-trivial evidence for the commit decision comes from one intracortical participant in a copy task. On that participant, a stream-level rule with a high-probability guarantee kept the wrong-commit rate below 5% when it was calibrated on earlier days and tested on later ones, and in random re-splits of the same data. It was faster than the same policy without commits by about 1% and 3%, which is within the uncertainty of the user model, and it was slower than always committing for the published decoder. On sEMG and MEG the layer reduced to always offering, the correct behaviour for decoders whose top sentence was rarely right; it cannot create information that the decoder lacks.

Which guarantee to deploy, and for which unit. CRC met its target as a mean, as it promises, but a third or more of single calibration sets exceeded α. LTT bounds the risk of the calibration distribution with high probability, at almost no speed cost against CRC with the max rule. That distribution is a mixture of sessions. On the published gate, one day in 14 ran at about 2α under LTT, and a user meets one session at a time. The deployable statement is therefore "at most 5% wrong commits over the mix of sessions that the calibration data represent", with its assumptions, its unit and the measured silent-error rate (Section S21). Per-session control needs per-session calibration: with LTT at least 45 calibration sentences per session, which is a long session-start block for a BCI user, or a group-conditional method that pools strength across sessions. In our time-ordered tests, a session-start block of half a day kept the pooled rate at 0.010–0.015 but did not remove the high day.

Who sets α and β. The marginal wrong-commit rate is the budget of harm per sentence attempted; the error among commits is what a user sees in committed text. With current decoders the two diverge: one in five commits of the guaranteed rule on our decoder was wrong, and a user who sees this will learn to check every commit, which turns commits into offers. A valid joint bound with $\beta = 0.10$ almost never committed with a few hundred calibration sentences; $\beta = 0.20$ was usable only on the published gate. Potential BCI users asked for at least 90% accuracy [12, 13]; they were asked about accuracy in general, and our reading that this suggests $\beta \leq 0.10$ for committed sentences is an interpretation. The targets should be chosen by the user with their clinician and communication partners, and may differ by context. Our exact-match loss treats a harmless near miss like a meaning-changing error; a loss weighted by semantic severity, scored by users, is the natural next target. Conformal selection with false-discovery-rate control [64, 65] bounds the expected error among commits and may commit more at the same β; it is the natural comparator to test next.

A commit rule needs an input-validity check. Calibration data contain real streams only, so a guarantee calibrated on them says nothing about streams that carry no neural information. Our plain threshold committed most uniform streams on our decoder, because its LM turns flat posteriors into fluent sentences with high calibrated confidence. The guard stopped this, but it did not lower wrong commits on real data, and with the full-fit pipeline it raised them. A deployed system needs a separate check that the input carries evidence, tested on signal-destroying controls.

Strong decoders still need a gate. The published decoder has a WER of 0.043, yet 19% of its sentences were wrong. Gating its text by agreement with a light reference decoder on the same stream kept 95% of the always-commit eWPM under the main user model (18.65 vs

19.61, fixed split A) and reduced wrong commits from 19% to 2.9%. This needs the stream and a reference decoder; the published text alone gave no usable confidence.

A frozen configuration for a confirmatory test. All analyses, including those that shaped the method, used the same Brain-to-Text '25 data, so all results here are exploratory. For a confirmatory test on data we have not used (a second participant, or new sessions), we fix this configuration: frame and sentence temperatures; the guard as in Table S5; the policy grid of Section S6 fitted on fit-A; LTT with the max rule on a disjoint calibration set that precedes the test data in time; $\alpha = 0.05$, $\delta = 0.1$; joint LTT with $\beta = 0.10$ and 0.20 as secondary rules. The primary outcomes are the pooled and per-session wrong-commit rates, the error among commits and the commit rate on the signal-destroying controls. This configuration is tagged `paper1-v4-frozen-config` in the repository; later commits changed only the manuscript text. It replaces the tag `paper1-v3-frozen-config` (commit `65359dc`), which froze the plain threshold without the guard; that configuration failed the no-information control (Section 6.6), so it must not be used. We keep the old tag for the record.

8 Limitations

1. **One implant participant** in a copy task, with online logits we cannot re-run, is the only non-trivial test of the commit decision. Our intervals cover days and splits of this person, not people. The time-ordered test has 10–13 test days, too few to confirm per-day control.
2. **The guarantee covers the mix of sessions**, not each session; one day ran at about 2α. All guarantees count only observed errors, and stream-level gating cannot detect intent mismatch.
3. **The input-validity check** that made the layer safe on no-information streams (the guard) did not lower wrong commits on real data (it raised them with the full-fit pipeline), and it was tested on synthetic controls only.
4. **Composed sEMG sentences and a weak MEG decoder**: isolated presses with oracle timing and templated test sentences (295 of 551 with a near variant in the LM text); 17 unique MEG test sentences, too few for a sentence-level guarantee.
5. **No live users.** Only two user costs are measured, in one gaze-typing user; speed rankings change across plausible variants; the word-completion baseline is weaker than clinical AAC; the loss is exact match. A user study needs ethics approval.

9 Conclusion

A trust layer on a per-frame output stream can decide whether a neural text-entry decoder commits, offers or abstains, with a high-probability bound on the wrong-commit rate. On one intracortical participant, Learn-then-Test with the max rule kept wrong commits below 5% when calibrated on earlier days and tested on later ones, but the bound covers the mix of sessions, not each session. About one in five (our decoder) or one in thirteen (published decoder) committed sentences was still wrong, and a valid bound on the error among commits almost never committed. Without an input-validity check, the layer committed no-information streams. Next steps are a confirmatory test of the frozen configuration on new data, per-session calibration, and a user study of which error rates users accept.

Ethics, data and code availability

This study is a secondary analysis of publicly available, de-identified data sets that their original investigators released under public licences: Brain-to-Text '25 (BrainGate2, ClinicalTrials.gov NCT00912041; CC0 [30]), Toronto sEMG typing (CC BY 4.0 [75]) and SpanishBCBL MEG (CC BY-NC 4.0 [76]). No new data were collected and no experiment with human participants was run. The original studies report their ethics procedures in their publications: the intracortical data come from participant T15 of the BrainGate2 clinical trial, collected under that trial's consent and institutional review board approval [2]; the MEG study was approved by the local ethics committee and its participants gave informed consent [6]; the sEMG data set and its collection are described in its data descriptor [9]. No further approval was required for this secondary analysis. From Brain-to-Text '25 we use only the logits, cue sentences and published sentences. We did not attempt re-identification of any participant. Quoted sentences are public copy-task cues, not the participant's own words. No raw recordings or participant data are redistributed (data sets and licences: Table S20).

The NeuralTextStream specification, conformance suite, reference implementation and all experiment scripts are at `https://github.com/zhangzibo/fidetype` (Apache-2.0), without weights trained on SpanishBCBL or copied Brain2Qwerty code. Every table and figure is generated from committed result files by committed scripts. The per-split item tables are derived numbers (no signals or texts) kept in a local cache; they can be rebuilt with the committed commands. The version of the code used for this paper is tagged `paper1-v4-frozen-config` in that repository. This article is distributed under the Creative Commons Attribution 4.0 International licence (CC BY 4.0). The trust layer took a median of 0.18–0.30 ms per decision on one CPU thread (95th percentile at most 1.06 ms), against 45–232 ms for the beam-search decoders.

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Competing interests

The author declares no competing interests.

Use of AI assistance

Code, analyses and manuscript drafting were carried out with substantial assistance from an AI system (Claude, Anthropic). The author directed the work, reviewed the results and takes full responsibility for the content. All numbers come from committed scripts and result files.

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A Trust decision and main user model

Algorithm 1 is the decision for one item; Algorithm S1 (supplement) gives the fitting and calibration. The reference implementation is `fidetype.trust` and `fidetype.eval.harness`. Table 5 lists the main simulated user; only the first two values are measured [27].

Algorithm 1 Trust decision for one item

Require: stream $\log p_t(v)$, $t = 1..F$; fitted τ_f, τ_s, q; policy $(\lambda, r_{\max}, r_{\text{abst}}, k_{\max}, \text{fallback})$

$\tilde{\ell}_t(v) \leftarrow \log p_t(v)/\tau_f - \log \sum_u p_t(u)^{1/\tau_f}$ ▷ frame calibration, before the LM

$(y_1..y_K, \text{score}_1..\text{score}_K) \leftarrow$ CTC prefix beam search with LM on $\tilde{\ell}$ ▷ K-best list

if the list is empty **then return** ABSTAIN

end if

$\hat{p}_j \leftarrow \text{softmax}_j(\text{score}/\tau_s)$; $\quad \hat{p}_{\text{top1}} \leftarrow \hat{p}_1$

$C \leftarrow \{\, j : 1 - \hat{p}_j \le q \,\}$, ordered by $\hat{p}_j$ ▷ conformal sentence set

for each character m of y_1 **do** ▷ acoustic confidence

  char streams: $a_m \leftarrow \max_{t \in \text{span}(m)} \exp \tilde{\ell}_t(y_{1,m})$ (CTC forced alignment of y_1)

  phoneme path: $a_m \leftarrow 0.5^e \min_{\text{phonemes of the word}} \exp \tilde{\ell}$ at their peak frames (e = lexicon edits)

end for

low $\leftarrow \frac{1}{|y_1|} \#\{m : a_m < 0.5\}$ ▷ low-evidence share

cons $\leftarrow \sum_{j \le 5} \hat{p}_j \,\text{sim}(y_1, y_j) / \sum_{j \le 5} \hat{p}_j$, $\quad \text{sim}(a,b) = 1 - \text{WED}(a,b)/\max(|a|_w, |b|_w)$

$g \leftarrow \min\big(1, \max(\text{low}, 1 - \text{cons})\big)$ ▷ guard risk; WED = word edit distance

if $g \ge r_{\text{abst}}$ **then return** ABSTAIN

end if

$s \leftarrow \hat{p}_{\text{top1}}$ if $g \le r_{\max}$ else $-\infty$ ▷ commit score

if $s \ge \lambda$ **then return** COMMIT y_1

end if

if $1 \le |C| \le k_{\max}$ and C does not contain the "other" option **then return** OFFER C

end if

return OFFER $y_1..y_{k_{\max}}$ if fallback = offer and $k_{\max} \ge 1$, else ABSTAIN

Table 5: Main simulated-user parameters. Variants: Section S4.

Parameter	Value	Basis
Fallback (gaze keyboard) speed	20 characters/min	measured, eye-gaze user LP1 [27]
Scan and select one of k word alternatives	$1.0 + 1.0 \log_2(k+1) + 1.5$ s (4.5 s for $k = 3$)	sum close to the measured 4.7 s [27]
Select one of k offered sentences	$1.0 + 0.5 \log_2(k+1)$ s	assumption
Reading time	0.3 s per word, 0.5 s per candidate	assumption
Navigate to and delete a wrong word	3.0 s (2 selections)	assumption
Miss probability of a wrong, unmarked word after a commit	0.05	assumption (no measurement)
Errors in offered candidates	always seen (main model)	assumption
Full retype overhead	1.5 s; chosen when cheaper than fixing words	assumption (rational user)
Abstain overhead	0.5 s, then fallback typing	assumption
Neural input speed, MEG / B2T '25 / sEMG	152 / 159.3 / 75.8 characters/min	device setting